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Studierichting: Informatica

Oefeningenreeks 4A nr. 43.12

$$\begin{aligned}
 & \int \frac{x^4 - 12x^2 + 1}{4x^5 - 4x^4 + x^3} dx \\
 & \Rightarrow 4x^5 - 4x^4 + x^3 = x^3 (4x^2 - 4x + 1) = x^3 (2x - 1)^2 \\
 & \Rightarrow \frac{x^4 - 12x^2 + 1}{x^3 (2x - 1)^2} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x^3} + \frac{D}{2x - 1} + \frac{E}{(2x - 1)^2} \\
 & C = \left. \frac{x^4 - 12x^2 + 1}{(2x - 1)^2} \right|_{x=0} = \frac{1}{1} = 1 \\
 & B = \left. \frac{\frac{x^4 - 12x^2 + 1}{(2x - 1)^2} - 1}{x} \right|_{x=0} = \left. \frac{x^4 - 12x^2 + 1 - 4x^2 + 4x - 1}{x (2x - 1)^2} \right|_{x=0} \\
 & = \left. \frac{x^3 - 16x + 4}{(2x - 1)^2} \right|_{x=0} = 4 \\
 & A = \left. \frac{\frac{x^3 - 16x + 4}{(2x - 1)^2} - 4}{x} \right|_{x=0} = \left. \frac{x^3 - 16x + 4 - 4(2x - 1)^2}{x (2x - 1)^2} \right|_{x=0} \\
 & = \left. \frac{x^3 - 16x + 4 - 16x^2 + 16x - 4}{x (2x - 1)^2} \right|_{x=0} \\
 & = \left. \frac{x^3 - 16x^2}{x (2x - 1)^2} \right|_{x=0} = \left. \frac{x^2 - 16x}{(2x - 1)^2} \right|_{x=0} = 0 \\
 & E = \left. \frac{x^4 - 12x^2 + 1}{x^3} \right|_{x=\frac{1}{2}} = \frac{\left(\frac{1}{2}\right)^4 - 12\left(\frac{1}{2}\right)^2 + 1}{\left(\frac{1}{2}\right)^3} = \frac{\frac{1}{16} - 3 + 1}{\frac{1}{8}} \\
 & = -\frac{31}{2} \\
 & D = \left. \frac{\frac{x^4 - 12x^2 + 1}{x^3} + \frac{31}{2}}{2x - 1} \right|_{x=\frac{1}{2}} = \left. \frac{\frac{2x^4 - 24x^2 + 2}{2x^3} + \frac{31x^3}{2x^3}}{2x - 1} \right|_{x=\frac{1}{2}} \\
 & = \left. \frac{2x^4 + 31x^3 - 24x^2 + 2}{2x^3(2x - 1)} \right|_{x=\frac{1}{2}} \\
 & = \left. \frac{(2x - 1)(x^3 + 16x^2 - 4x - 2)}{2x^3(2x - 1)} \right|_{x=\frac{1}{2}}
 \end{aligned}$$

$$\begin{aligned}
&= \frac{x^3 + 16x^2 - 4x - 2}{2x^3} \Big|_{x=\frac{1}{2}} = \frac{1}{2} \\
&\Rightarrow \int \left(\frac{4}{x^2} + \frac{1}{x^3} + \frac{1}{4x-2} + \frac{-31}{2(2x-1)^2} \right) dx \\
&\quad \boxed{= \frac{-4}{x} - \frac{1}{2x^2} + \frac{1}{4} \ln |4x-2| + \frac{31}{4(2x-1)} + k}
\end{aligned}$$

References